Deepfake Voice Detection

Milestone 2 | Dataset Identification & Preparation

What we collected, what we found, and how we will evaluate fairly.



PROJECT GOAL

What must the dataset let us test?

Detect AI-generated speech across Indian languages—and check whether the detector generalizes.

Real speech

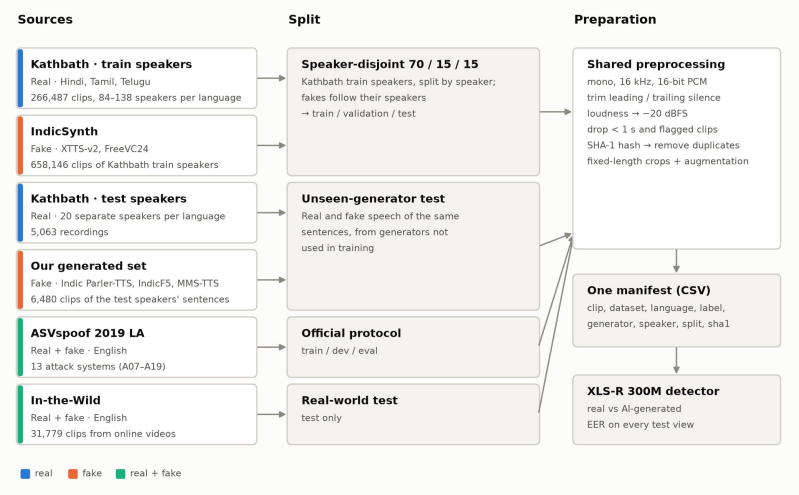
Kathbath gives real Hindi, Tamil and Telugu audio.

Synthetic speech

IndicSynth provides two Indian-language generation methods.

Unseen conditions

Extra generators and English datasets test generalization.



Core principle: keep speakers and test sources separate so results reflect voice authenticity—not who spoke or which tool generated it.

DATASETS

Five datasets, five different roles

A combined dataset portfolio covers real speech, synthetic speech, and three types of evaluation.

Kathbath

266,487 real clips
Hindi • Tamil • Telugu

IndicSynth

658,146 metadata clips
XTTS-v2 • FreeVC24

ASVspoof 2019 LA

71,237 evaluation clips
English benchmark

In-the-Wild

31,779 clips
English real-world test

Group-generated set

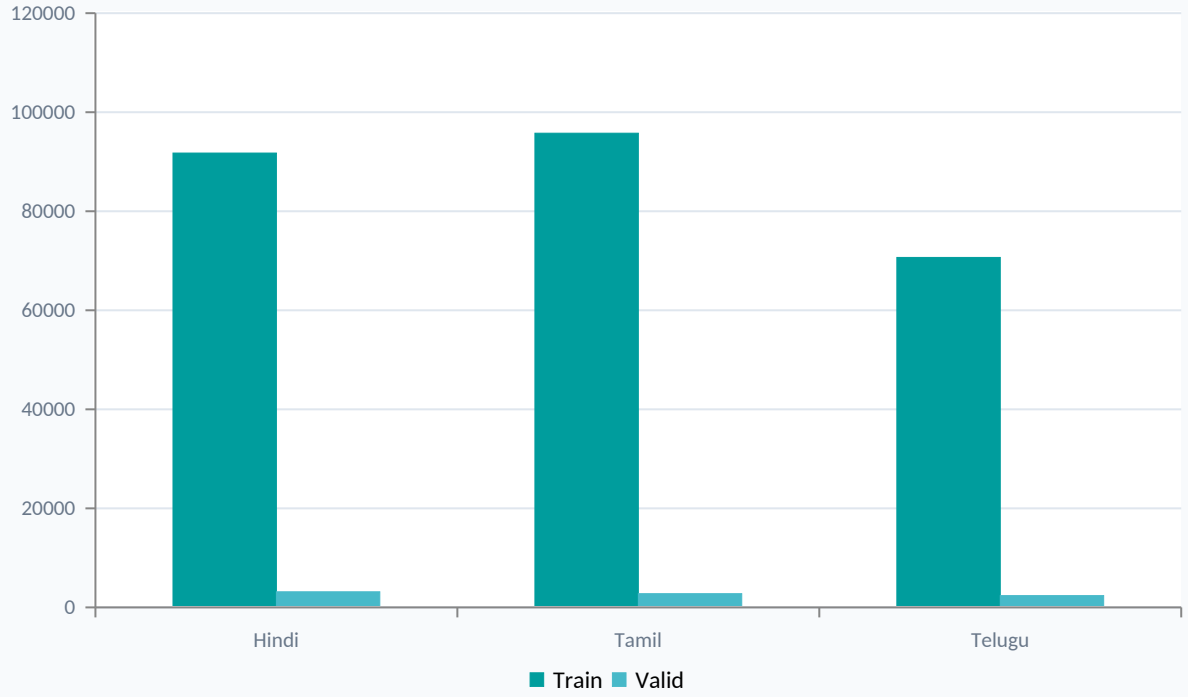
6,480 clips • 11.25 hours
3 additional generators

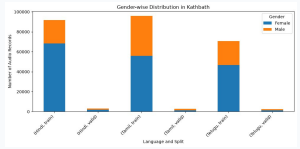
Note: IndicSynth is a metadata inventory (3,000 clips sampled for audio EDA); ASVspoof is its evaluation partition. These counts are not a quality ranking.

Kathbath: the real-speech foundation

All 266,487 loaded clips decoded successfully; the native validation split is not speaker-independent.

Clips by language





94,903
Hindi total

98,532
Tamil total

73,052
Telugu total

0
Decode errors
266,487 checked

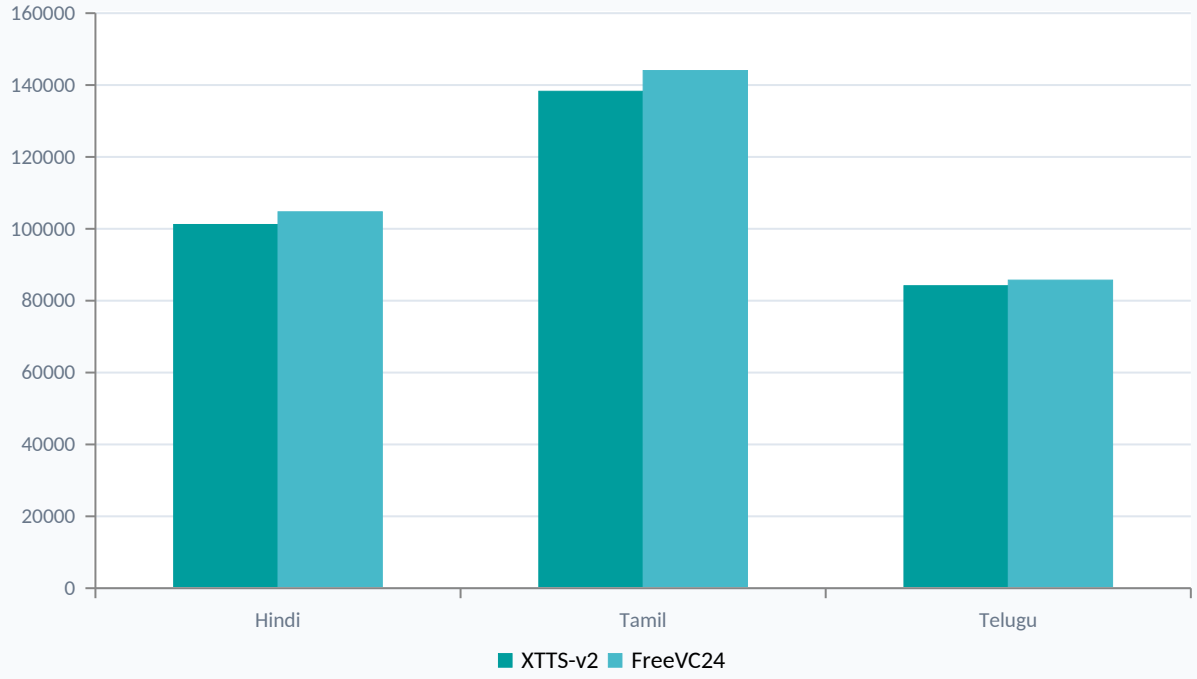
Key caution

All 20 validation speakers per language also appear in train.

IndicSynth: a large fake-speech pool, with an imbalance

The metadata is complete; acoustic EDA decoded a 3,000-clip sample.

Clips by language and generator



323,646
XTTS-v2

334,500
FreeVC24

≈1,260
Estimated hours

23%
Tamil female voices
32 F / 106 M speakers

Split implication

All source and target speakers come from Kathbath training speakers.

English datasets reveal different evaluation risks

Keep each dataset’s protocol and class balance visible.

ASVspoof 2019 LA

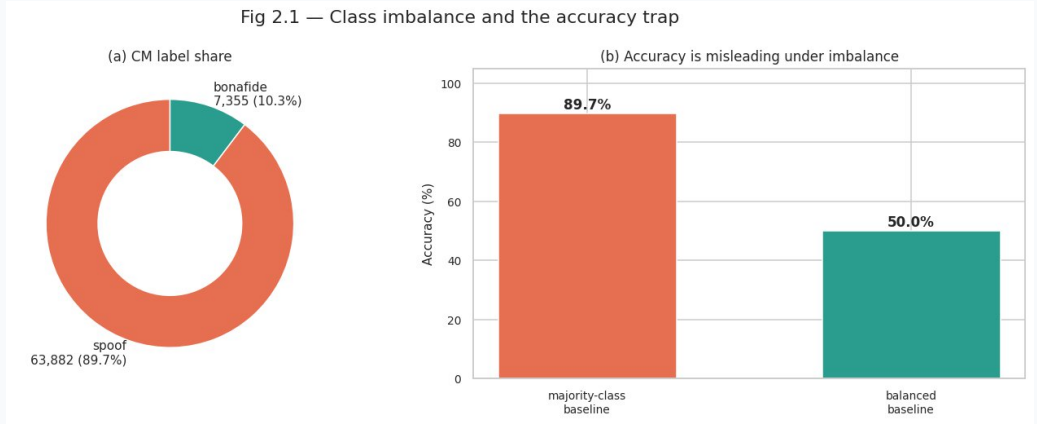
71,237 clips in eval
7,355 real • 63,882 spoof
13 attack systems

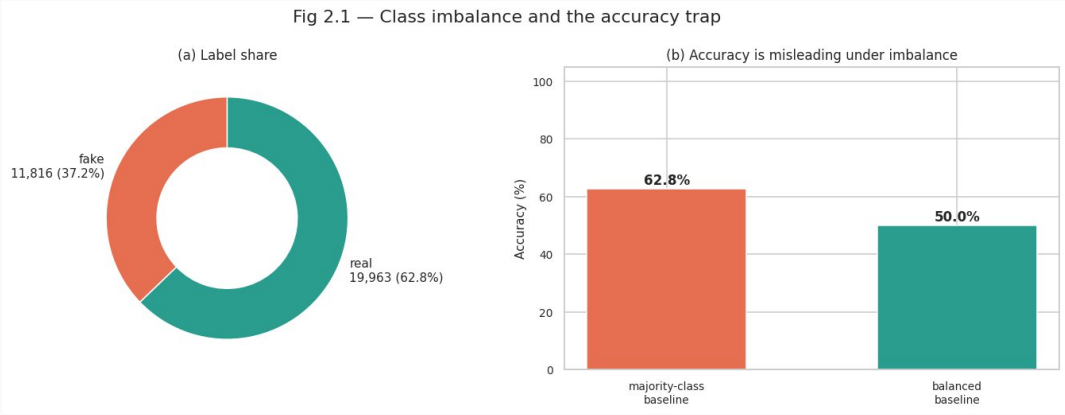
In-the-Wild

31,779 clips
19,963 real • 11,816 fake
54 speakers in metadata

Why it matters

ASVspoof is 89.7% spoof: a model always saying “spoof” gets 89.7% accuracy.



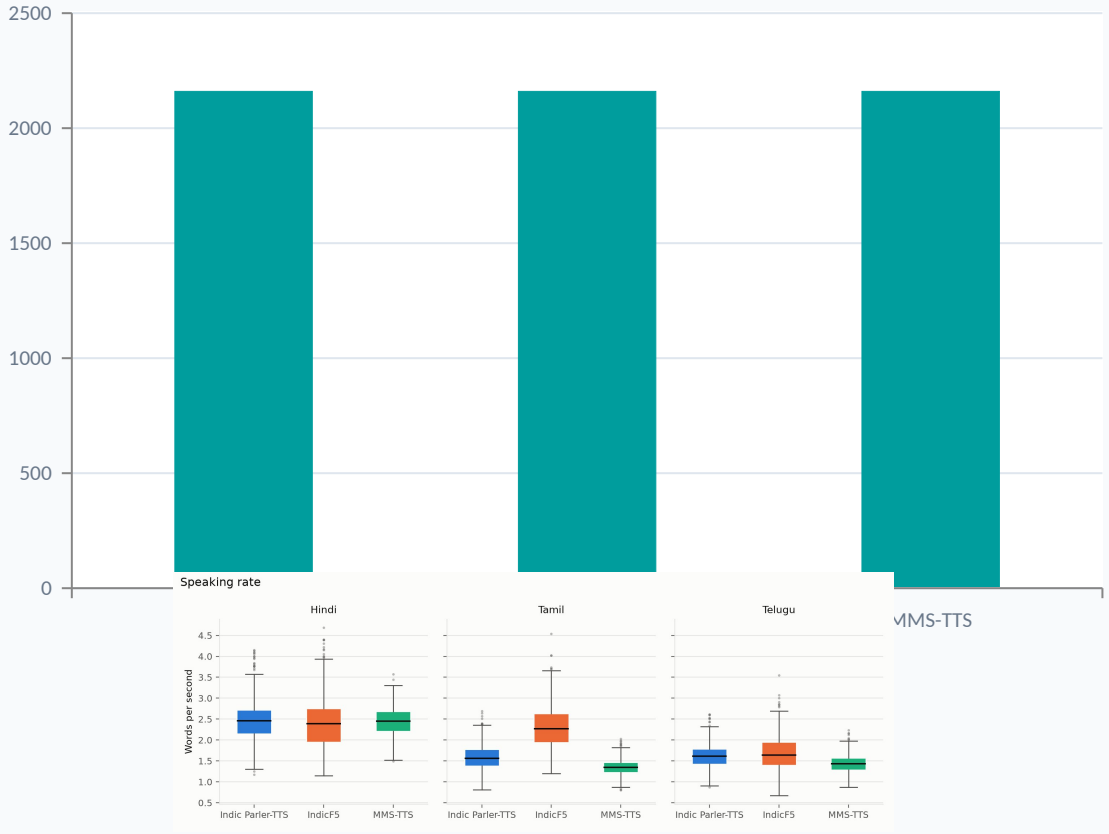


Use EER and per-class recall alongside accuracy. Official ASVspoof train/dev audio still needs to be obtained.

Our generated set tests three unseen generators

Same 720 sentences per language are spoken by each generator, controlling text when comparing voices.

Generated clips per model

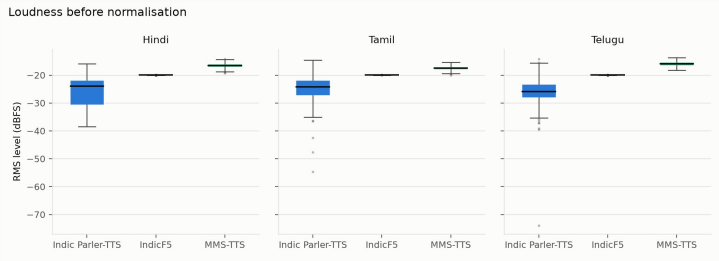


6,480
Total clips

11.25 h
Audio duration

3
Languages
Hindi • Tamil • Telugu

0
Missing clips
All expected clips present



Native sample rates differ: 44.1 kHz • 24 kHz • 16 kHz.

QUALITY FINDINGS

Data checks passed; audio differences still matter

No broken files or missing labels were found in the audited data. Some clips are flagged for review.

0
Kathbath decode errors

0
Missing metadata fields*

582
Kathbath clips outside 2–15 s

≈12k
Clipping flags to review

Sample rate / format

Sources use 16, 24 or 44.1 kHz; formats include 16-bit PCM and 32-bit float.

Loudness / silence

Levels and leading/trailing silence differ by source—and can correlate with labels.

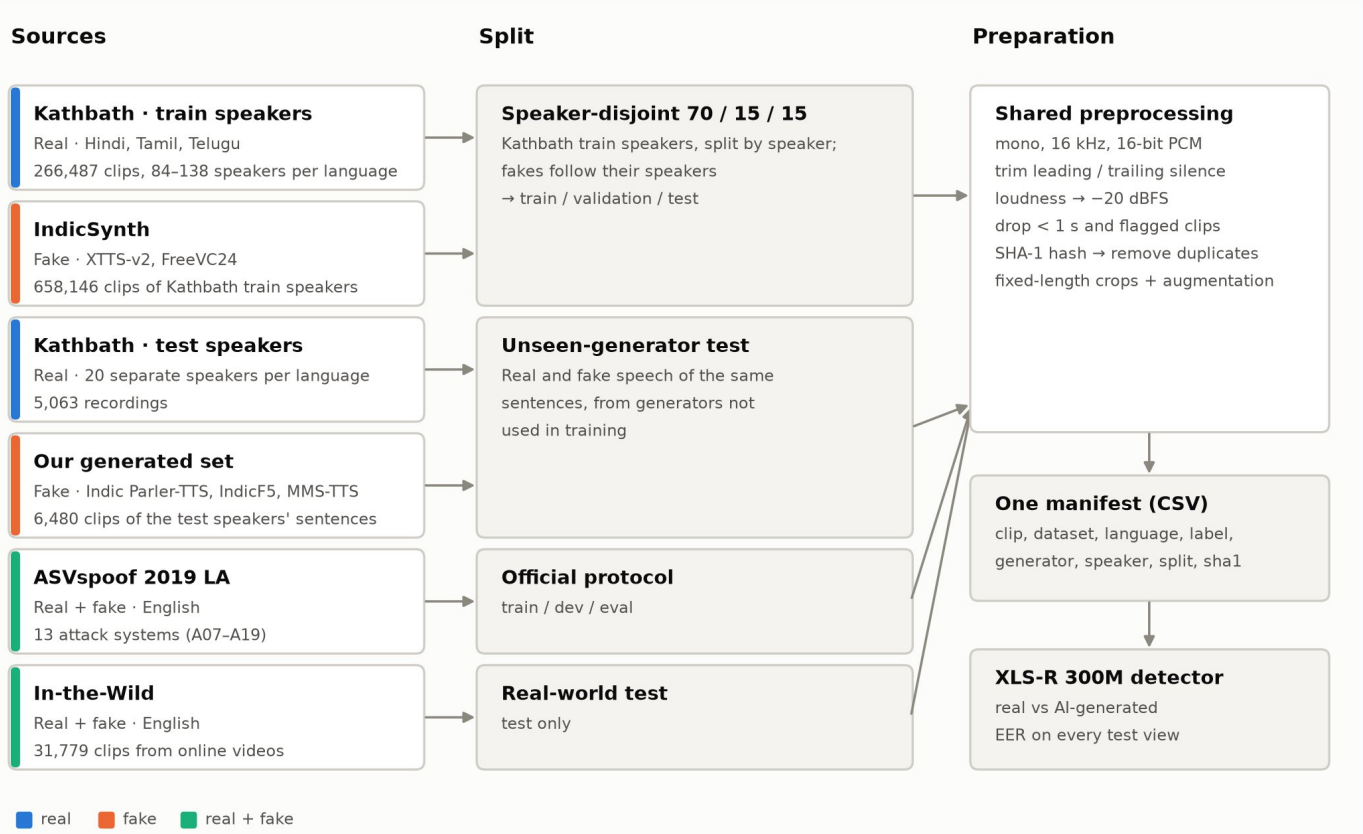
One silent clip

Generated clip `te_0623` is flagged and excluded through the manifest.

**For IndicSynth, empty fields are expected where a generator does not use that field. Some decode checks were sampled, not run on every source clip.*

The split must prevent the model from memorizing speakers

IndicSynth uses Kathbath training speakers, so split by speaker—not by clip.



Training / validation

Kathbath training speakers + their IndicSynth fakes; split by speaker.

Unseen-speaker test

Kathbath test speakers supply real speech only; generated test uses their sentences.

External tests

Keep ASVspoof's official protocol and In-the-Wild separate.

Also check FreeVC source/target links, sentence overlap and duplicate audio hashes after preprocessing.

PREPROCESSING

One preprocessing pipeline for every source

The goal is to remove differences in format, loudness and silence that could give away the dataset instead of the voice.

Decode
Read audio

›

Standardize
Mono • 16 kHz

›

Trim
Edge silence

›

Normalize
−20 dBFS target

›

Check
Drop flagged/short

›

Manifest
Labels + SHA-1

Training augmentation

Noise, codecs and phone-channel effects; apply equally to real and fake.

Status

Pipeline is documented; final end-to-end processed manifests and model results remain future work.

READINESS

Coverage is strong; two gaps remain

The dataset is adequate for the planned experiments, but it does not cover every real-world condition.

Good coverage

Real + fake Indian-language speech; 3 held-out generators; two English evaluation sources.

Gap 1: generator variety

Only XTTS-v2 and FreeVC24 are in the Indian-language synthetic training pool.

Gap 2: phone audio

No native phone-quality dataset; plan to simulate it through training augmentation.

How we should judge the detector

EER

Useful for spoof detection with imbalanced classes.

Per-class recall

Check both real and fake speech—not just overall accuracy.

Subgroup results

Report by language, gender and generator.

NEXT STEPS

Next: turn the data plan into model-ready splits

Finish reproducibility and leakage checks before interpreting model performance.

1. **Get official ASVspoof train/dev audio** — Keep the benchmark's official protocol.
2. **Run preprocessing end-to-end** — Save processed audio, manifest, hashes and summary stats.
3. **Verify leakage checks** — Speakers, FreeVC links, text overlap and duplicate hashes.
4. **Finalize balanced samples** — Confirm language/gender counts and split integrity.
5. **Train baseline models** — Report EER, per-class recall and held-out-generator results.

Milestone 3 exit: versioned manifest + split audit + preprocessing validation + baseline-ready partitions.